

Seat No.

Total number of printed pages: 2

[25]

SARDAR PATEL UNIVERSITY

MASTER OF SCIENCE (INFORMATION TECHNOLOGY)

SEMESTER - II

COURSE PS02EINT03 (TRENDS IN ICT)

Friday April 21st 2017

Time: 2:00 pm to 5:00 pm

Max. Marks: 70

Q-1 Choose the most appropriate option for each question:

[8]

- Which of the following is *not* an advantage of a distributed system?
 - Economics
 - Inherent distribution
 - Incremental growth
 - None of these
- What is the full form of SAP?
 - Software Applications Products
 - Systems Applications Products
 - Software Architecture Products
 - Systems Architecture Products
- Which of the following adds domain specific classes in J2ME?
 - A profile
 - A configuration
 - A package
 - None of these
- Which among the following has the highest speed?
 - DDR2 RAM
 - DDR3 RAM
 - L2 cache
 - L3 cache
- Which notation is used to write IPv6 addresses?
 - Dotted-decimal notation
 - Dotted-hexadecimal notation
 - Decimal notation
 - Hexadecimal notation
- What is the basic unit of a Bluetooth networking system?
 - nanonet
 - piconet
 - femtonet
 - micronet
- What is the full form of XML?
 - Extended Markup Language
 - Expanded Markup Language
 - Extensible Markup Language
 - Expandable Markup Language
- Which of the following is placed between the internal networks of an organization and the external networks to provide security?
 - A firewall
 - A switch
 - A hub
 - A router

Q-2 Answer the following questions (Any Seven):

[14]

1. Define different types of real time systems.
2. List any four characteristics of an operating system.
3. List any four advantages of client-side scripting.
4. Explain important characteristics of the CPU.
5. Differentiate between authentication and authorization.
6. Explain in brief types of firewalls.
7. Explain the servlet lifecycle in brief.

8. Write features of J2EE.
9. What is a substitution cypher? Give a simple example.

Q-3 Answer the following questions:

- A. Explain with a figure the taxonomy of distributed and parallel systems given by Flynn. [6]
B. Explain tightly-coupled systems and loosely-coupled systems. Also describe bus-based architecture and switch-based architecture. Draw a figure illustrating their combinations. [6]

OR

- B. Explain the use and features of application servers. Describe the 3-tier client-server architecture. [6]

Q-4 Answer the following questions:

- A. What are web services? Explain the key protocols used for developing web services. [6]
B. Write a short note on SAP. [6]

OR

- B. Explain any three important hardware components of the computer system. [6]

Q-5 Answer the following questions:

- A. Write a short note on IPv6. [6]
B. Write a short note on Bluetooth. [6]

OR

- B. Write a short note on J2ME. [6]

Q-6 Answer the following questions:

- A. Explain *any one* in detail: i. XML ii. Servlets [6]
B. Write a short note on cryptography. [6]

OR

- B. Write a short note on different types of malware (computer virus). [6]
